CS532 Quiz 7: Surface Estimation and PointClouds

Name: _____

Question 1 of 2. How would you use normal estimation to improve ICP?

Solution: If we have normals estimated as well, we can do point to plane ICP instead of point to point ICP. That is, for each 3D point correspondence $p_i \leftrightarrow p_j$, use the target normal n_j and minimize the following point to plane distance rather than the full euclidean distance:

$$\sum_i (n_j^\intercal (Rp_i + t - p_j))^2$$

This uses the local tangent plane implied by the normal, gives faster convergence (especially on large smooth surfaces), and is less sensitive to tangential noise.

Question 2 of 2. Explain WHY each of the following would (or would not) break the ICP operating assumptions

1. Spatially varying rigid motion between point clouds
2. Different 3D point density between point clouds
3. Rotation-only motion

Solution:

1. This breaks ICP. Here different parts of the object move with different rigid transforms. No single (R, t) can align all points simultaneously, so the core ICP model (one global rigid transform explains both clouds) is violated.
2. A single rigid transform still exists, but ICP's least-squares objective implicitly assumes roughly uniform sampling so each surface region has similar influence. With very different densities, dense regions dominate the fit and sparse regions are effectively down-weighted. So the rigid-motion assumption still holds, but the uniform sampling assumption of ICP is weakened, which can bias or slow convergence rather than outright break ICP.
3. Pure rotation is just a special case of a rigid transform. So this does not break ICP